

King Abdullah University of
Science and Technology



جامعة الملك عبد الله
للعلوم والتقنية

Approximate Bayesian inference for Structural Equation Models (SEM)

The INLA approach

Haziq Jamil 

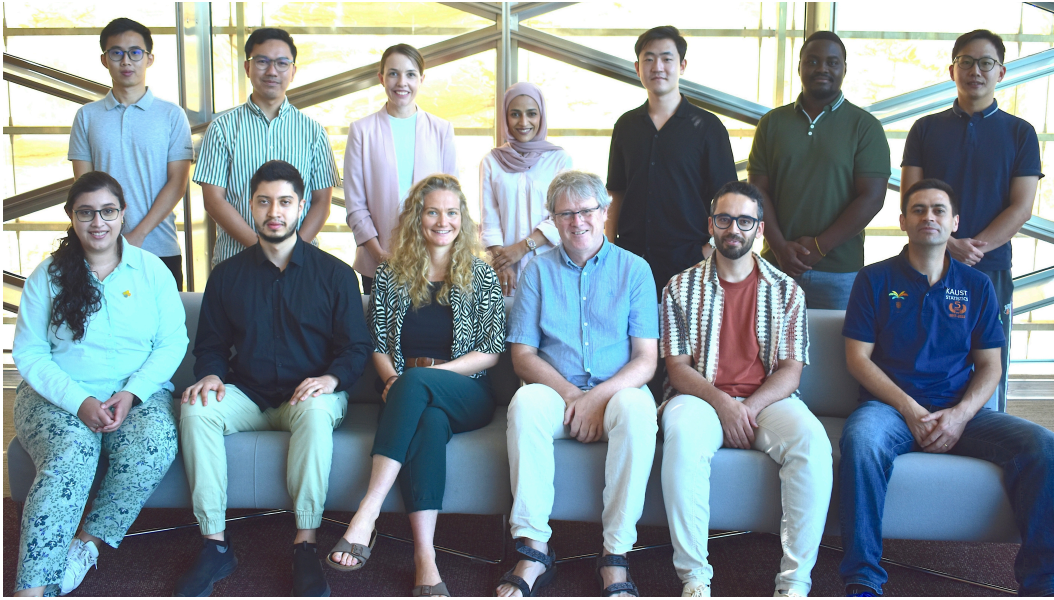
Research Specialist

BAYESCOMP @ CEMSE-KAUST

<https://haziqj.ml/kaust-gradsem> |  PDF

October 8, 2026





Bayesian Computational Statistics and Modeling Research Group (BAYESCOMP)

Led by Prof. Håvard Rue—*Integrated Nested Laplace Approximation (INLA, Rue et al., 2009)*

Foundations & priors

- Information deletion / unlearning
- PC priors & regularisation
- Informative correlation priors

Model development

- Latent variables: SEM, CFA, IRT
- Spatio-temporal / point processes
- Joint longitudinal–survival models

Inference & assessment

- INLA & variational Bayes
- Cross-validation & prediction
- Model checking & robustness

Scalable computation

- Sparse solvers
- Automatic differentiation
- GPU & distributed computing

APPLICATIONS

Education **fMRI** **Healthcare** **Ecology** **Climate** Air pollution Coral reefs
Psychology Neuroscience Public health **Epidemiology** **Environment** Conservation
spatial temporal spatiotemporal HIV Cancer dolphins Disease mapping Infectious diseases



Data context

Structural equation models (SEM) in a nutshell

Analyse multivariate data $\mathbf{y} = (y_1, \dots, y_p)^\top$ to **measure** and **relate** hidden variables $\boldsymbol{\eta} = (\eta_1, \dots, \eta_q)^\top$, $q \ll p$, and uncover complex patterns.

In the social sciences, latent variables are used to represent **constructs**—the *theoretical, unobserved* concepts of interest.



(Psychology)
Personality traits



(Healthcare)
Quality of life



(Political science)
Social trust



(Education)
Competencies

Photo credits: Unsplash @dtravisphd, @impulsq, @ev, @benmullins.



European Social Survey (ESS) Round 5

Total $n = 52,458$ respondents from 27 countries. Example of items measuring the construct “*Trust in Police Effectiveness*”:

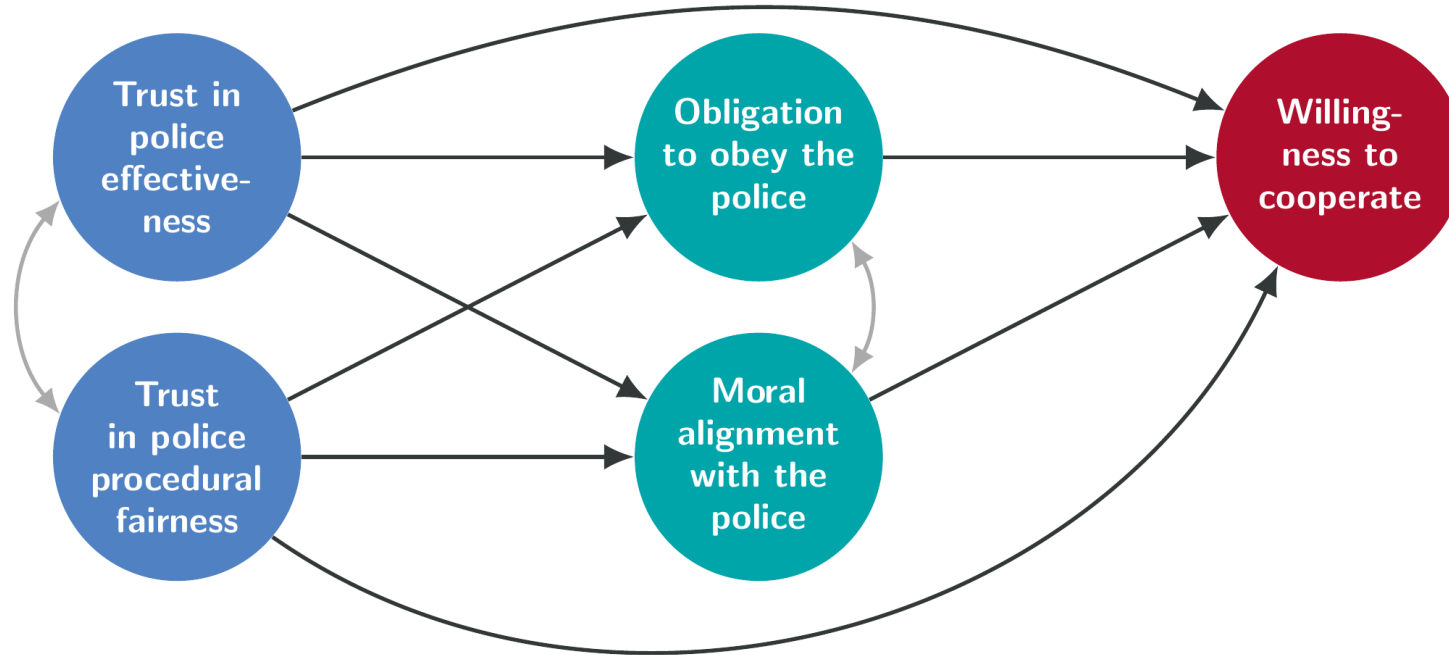
Ser.	Question
D12	Based on what you have heard or your own experience how successful do you think the police are at preventing crimes in [country] where violence is used or threatened?
D13	How successful do you think the police are at catching people who commit house burglaries in [country]?
D14	If a violent crime were to occur near to where you live and the police were called, how slowly or quickly do you think they would arrive at the scene?



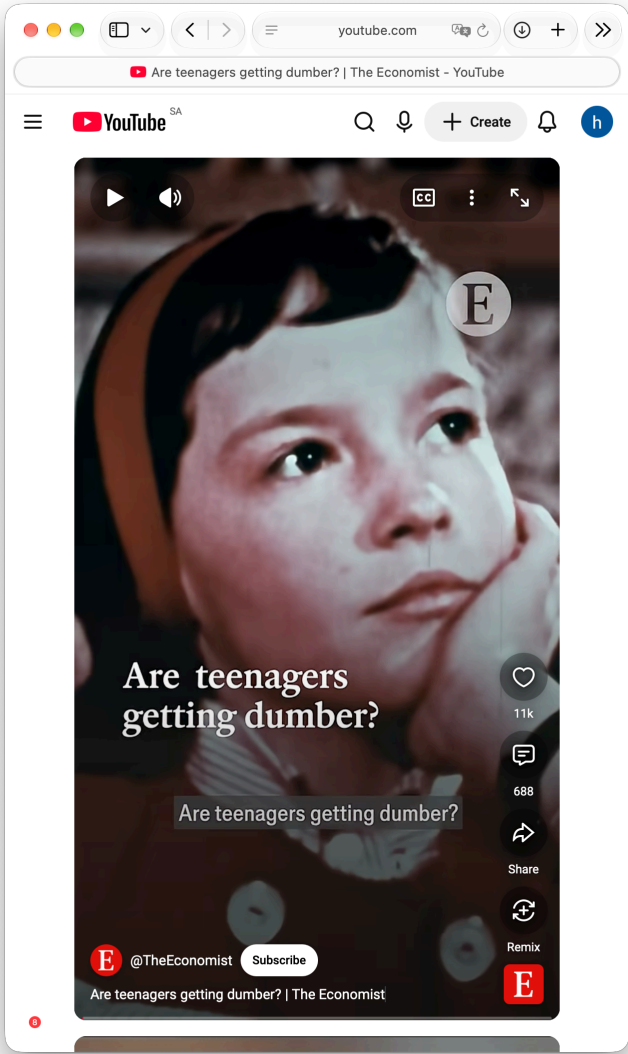
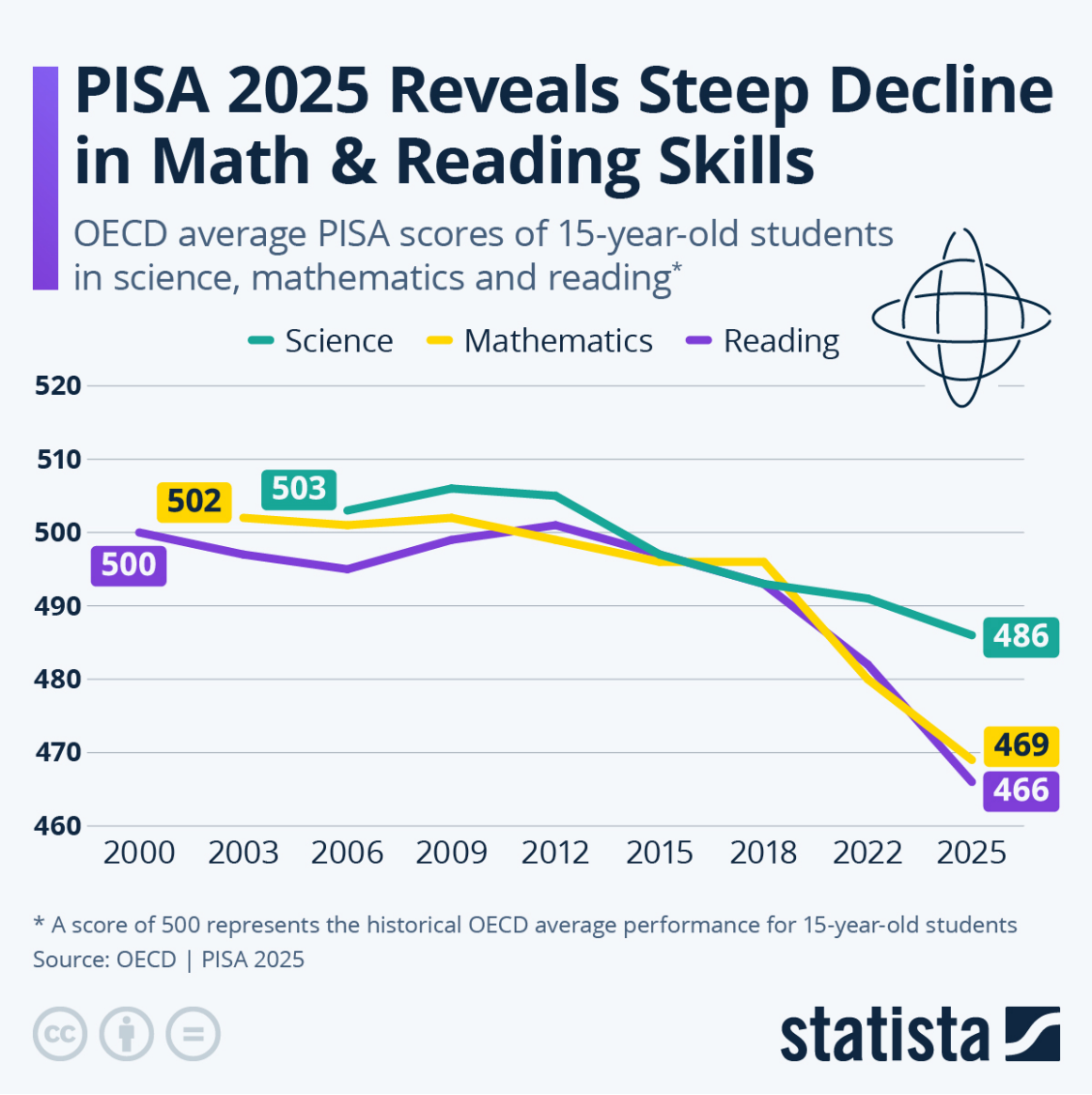
Data available from European Social Survey European Research Infrastructure (ESS ERIC) (2023).



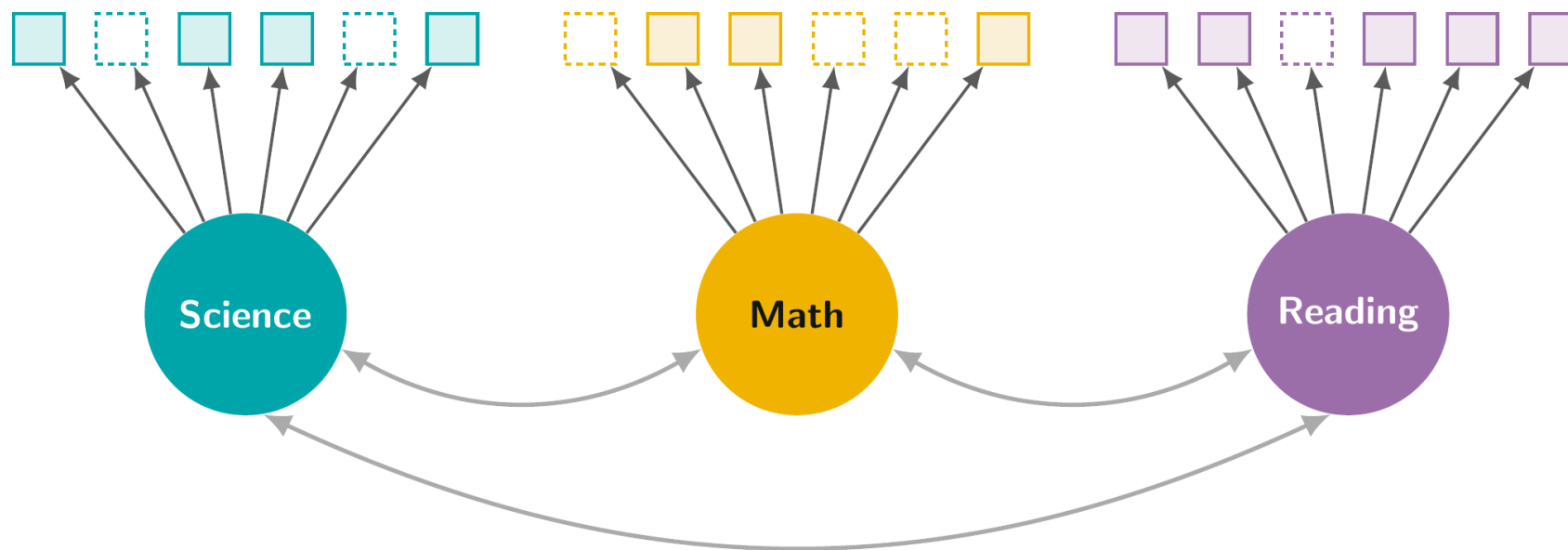
European Social Survey (ESS) Round 5 (cont.)



- Modelling procedural fairness theory linking trust, legitimacy, and compliance with authorities (Jackson, Bradford, et al., 2011; Jackson, Hough, et al., 2011).
- Big and layered data suited for Bayesian approaches. But model fit and evaluation can be computationally challenging or time-consuming.
- Often, country-level effects are ignored, but they can be crucial in understanding the variability in the latent constructs across countries.



PISA (cont.)



- PISA uses 2PL *item response theory (IRT)* models to measure unobserved proficiency from item responses.
- **Estimation involves large-scale, multi-group calibration and latent population modelling**, with incomplete forms and complex survey sampling.
- Population inference uses *plausible values* (draws from students' conditional proficiency posteriors), given fitted item and population parameters.



Advancing scalable Bayesian methods for latent variable modelling

Håvard Rue (PI) + Haziq Jamil (co-I)

- **Jamil, H.**, & Rue, H. (2026a). *Approximate Bayesian inference for structural equation models using integrated nested Laplace approximations* ([arXiv:2603.25690](#) [stat.ME]).¹
- **Jamil, H.**, & Rue, H. (2026b). *Implementation and workflows for INLA-based approximate Bayesian structural equation modelling* ([arXiv:2604.00671](#) [stat.CO]).¹
- Alhyari, M., **Jamil, H.**, Montcho, H., & Rue, H. (2026). *Deterministic Leave-One-Cluster-Out Cross-Validation for Multilevel Bayesian Structural Equation Models* ([arXiv:2609.00670](#) [stat.ME]).¹
- R Package: <https://github.com/haziqj/INLAvaan>

Supported by the 2025 KAUST Competitive Research Grant (CRG)².

WP1 Theory & model mapping

Fast Bayesian SEM for continuous, binary & ordinal items

WP2 Latent traits on a map

Spatial SEM & IRT with SPDE random fields

WP3 Latent traits in time

Growth curves & dynamic SEM

WP4 Software & dissemination

Open-source R package: INLAvaan

1. Under review at Psychometrika (Q1, Applied Mathematics).

2. <https://bayescomp.kaust.edu.sa/articles/2026/03/10/bayescomp-group-secures-crg-funding-revolutionize-latent-variable-modeling>



Outline



1. Brief overview of SEMs

- A running example
- Bayesian inference
- Marginal likelihood

2. INLA-based inference

- Joint Gaussian approximation with variational mean correction
- Skew-normal Laplace profiling of posterior marginals
- Gaussian copula sampling

3. The INLAvaan R package

Latent variables related to KAUST's agenda



FOOD

Cafiero et al. (2018)

The UN food insecurity measure

A Rasch model (1PL IRT) on yes/no questions, from worrying about food to going a whole day without eating; SDG 2.1.2.



WATER

Young et al. (2019)

Household water insecurity measure

Twelve questions, validated on 8,127 households at 28 sites in 23 countries, and unidimensional in factor analyses.



AI

Burnell et al. (2023)

LLMs have latent abilities too

Factor analysis reveals abilities like reasoning and comprehension. IRT used to find the most informative questions.



ENVIRONMENT

Grace et al. (2016)

SEM sees what raw correlations hide

Across 1,126 grassland plots on 5 continents, species richness boosts productivity, in contrast with data patterns.

Photo credits: Unsplash @pazarando, @_mrjn_esf, @katjaano, @jarlock55.





Structural equation models



A running example

Glycemic control and kidney health

Does poorer glycemic control lead to greater severity of kidney disease?

Observe $p = 6$ variables for each patient:

	Indicator	Description	Unit
y_1	HbA1c	3-month avg. blood glucose	%
y_2	FPG	Fasting plasma glucose	mmol/L
y_3	Insulin	Fasting insulin level	$\mu\text{U/mL}$
y_4	PCr	Plasma creatinine	$\mu\text{mol/L}$
y_5	ACR	Albumin-creatinine ratio	mg/g
y_6	BUN	Blood urea nitrogen	mmol/L

“*piecemeal*” thinking leads to

$$y_4 = \beta_0^{(4)} + \beta_1^{(4)} y_1 + \beta_2^{(4)} y_2 + \beta_3^{(4)} y_3 + \epsilon^{(4)}$$

$$y_5 = \beta_0^{(5)} + \beta_1^{(5)} y_1 + \beta_2^{(5)} y_2 + \beta_3^{(5)} y_3 + \epsilon^{(5)}$$

$$y_6 = \beta_0^{(6)} + \beta_1^{(6)} y_1 + \beta_2^{(6)} y_2 + \beta_3^{(6)} y_3 + \epsilon^{(6)}$$

- Does not give a clear and direct answer.
- Moreover, variables are assumed to be measured without error.

Example adapted from Song & Lee (2012).



Covariance-based approach

Sample correlation matrix looks like this¹:

	y ₁	y ₂	y ₃	y ₄	y ₅	y ₆
y ₁	1	0.82	0.77	0.44	0.48	0.43
y ₂	0.82	1	0.81	0.4	0.44	0.41
y ₃	0.77	0.81	1	0.42	0.44	0.43
y ₄	0.44	0.4	0.42	1	0.81	0.78
y ₅	0.48	0.44	0.44	0.81	1	0.82
y ₆	0.43	0.41	0.43	0.78	0.82	1

- The data suggests clustering of variables
 - y_1, y_2, y_3 measure *glycemic control* (**GlyCon**)
 - y_4, y_5, y_6 measure *kidney health* (**KdnHl t**)

- Easier to hypothesize relationships, e.g.

$$\mathbf{KdnHl t} = \alpha + \beta \mathbf{GlyCon} + \mathbf{error}$$

- But these are unobserved—we only see their *footprint* on the correlation structure:
 - Items in a cluster share a latent cause, and
 - Cross-cluster correlation flows through β .

- SEM models the **covariance structure**

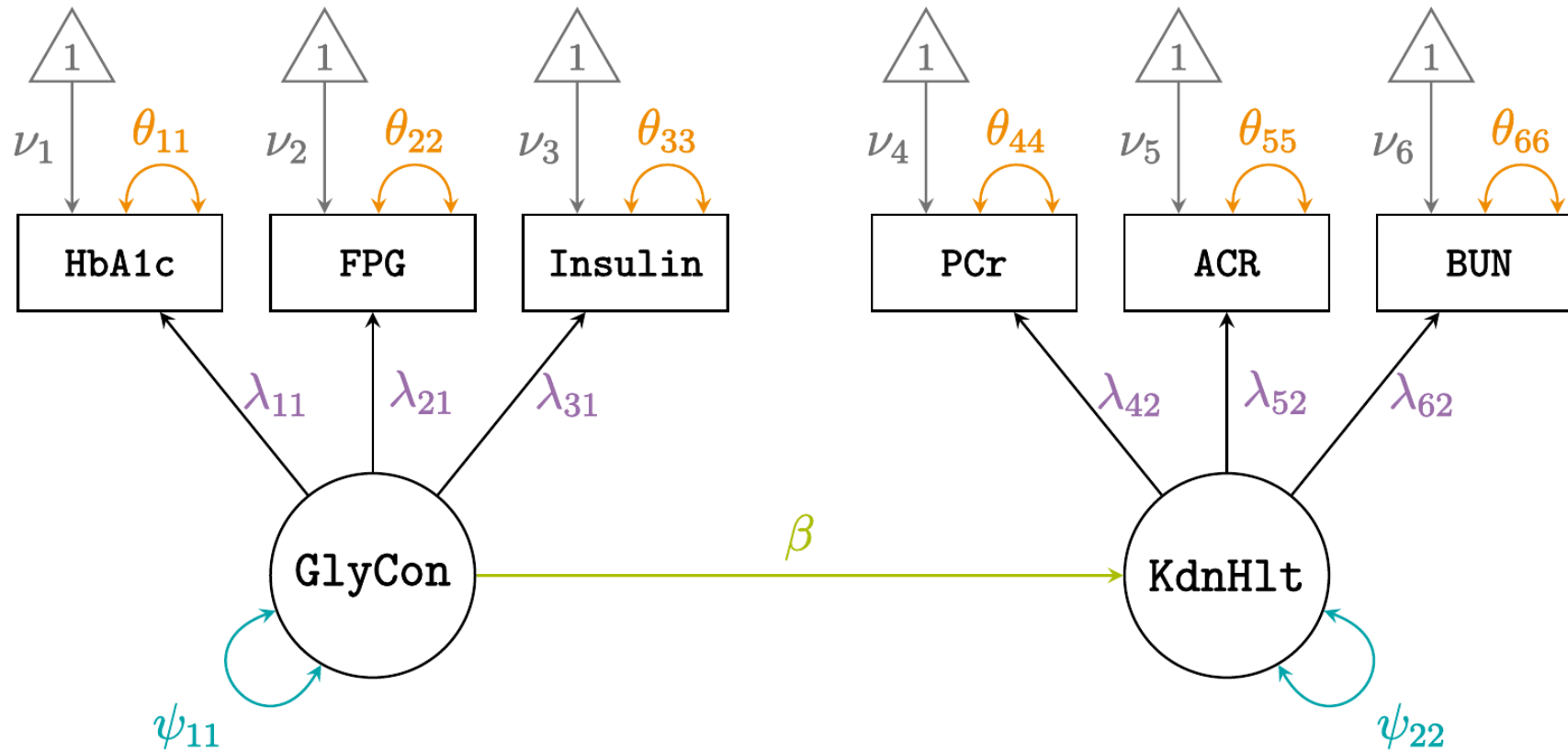
$$\Sigma = \Sigma(\vartheta),$$

where ϑ are model parameters.

¹. Simulated data from a two-factor SEM ($n = 250$).



Directed acyclic graph (DAG)



- **Measurement model:** loadings λ , intercepts ν , residual (co)variances θ .
- **Structural model:** regression coefficients β , latent (co)variances ψ .
- Collect free parameters in $\boldsymbol{\vartheta} \in \mathbb{R}^m$, where $m < p(p+1)/2 + p$.



SEM equations

$$\begin{pmatrix} y_1 \\ y_2 \\ y_3 \\ y_4 \\ y_5 \\ y_6 \end{pmatrix} = \begin{pmatrix} \nu_1 \\ \nu_2 \\ \nu_3 \\ \nu_4 \\ \nu_5 \\ \nu_6 \end{pmatrix} + \begin{pmatrix} \lambda_{11} & 0 \\ \lambda_{21} & 0 \\ \lambda_{31} & 0 \\ 0 & \lambda_{42} \\ 0 & \lambda_{52} \\ 0 & \lambda_{62} \end{pmatrix} \begin{pmatrix} \eta_1 \\ \eta_2 \end{pmatrix} + \begin{pmatrix} \epsilon_1 \\ \epsilon_2 \\ \epsilon_3 \\ \epsilon_4 \\ \epsilon_5 \\ \epsilon_6 \end{pmatrix}$$

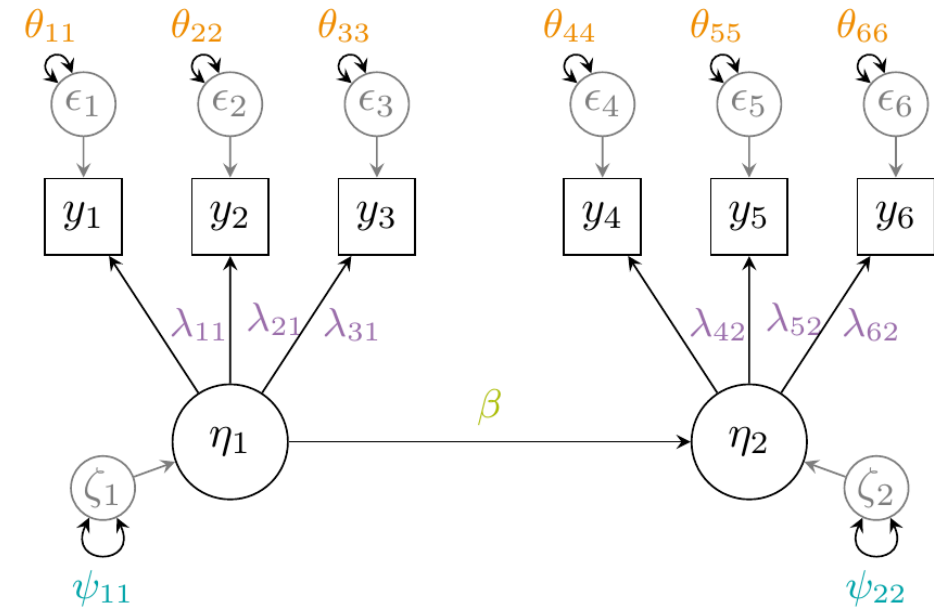
$$\begin{pmatrix} \eta_1 \\ \eta_2 \end{pmatrix} = \begin{pmatrix} \alpha_1 \\ \alpha_2 \end{pmatrix} + \begin{pmatrix} 0 & 0 \\ \beta & 0 \end{pmatrix} \begin{pmatrix} \eta_1 \\ \eta_2 \end{pmatrix} + \begin{pmatrix} \zeta_1 \\ \zeta_2 \end{pmatrix}$$

Or, more compactly as

$$\mathbf{y} = \boldsymbol{\nu} + \boldsymbol{\Lambda}\boldsymbol{\eta} + \boldsymbol{\epsilon}$$

$$\boldsymbol{\eta} = \boldsymbol{\alpha} + \mathbf{B}\boldsymbol{\eta} + \boldsymbol{\zeta}$$

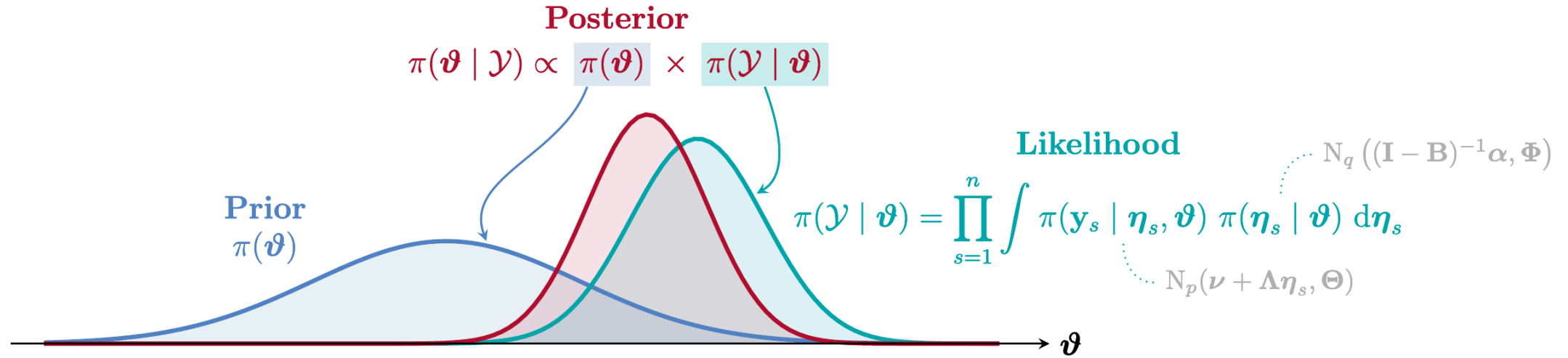
with assumptions $\boldsymbol{\epsilon} \sim \mathbf{N}_p(\mathbf{0}, \boldsymbol{\Theta})$,
 $\boldsymbol{\zeta} \sim \mathbf{N}_q(\mathbf{0}, \boldsymbol{\Psi})$, and $\text{Cov}(\boldsymbol{\epsilon}, \boldsymbol{\zeta}) = \mathbf{0}$.



- NB: Per person structure, assumed independent for each $s = 1, \dots, n$.
- Additionally, assume prior distributions
 - $\nu_i \sim \mathbf{N}(0, 32^2)$, $\alpha_j \sim \mathbf{N}(0, 10^2)$
 - $\lambda_{ij}, \beta \sim \mathbf{N}(0, 10^2)$
 - $\sqrt{\theta_{ii}}, \sqrt{\psi_{jj}} \sim \Gamma(1, 0.5)$, i.e. on the SDs



MCMC-based estimation



Data augmentation Gibbs

$$\pi(\boldsymbol{\vartheta}, \boldsymbol{\eta} | \mathcal{Y}) \propto \pi(\boldsymbol{\vartheta}) \prod_{s=1}^n \pi(\mathbf{y}_s | \boldsymbol{\eta}_s, \boldsymbol{\vartheta}) \pi(\boldsymbol{\eta}_s | \boldsymbol{\vartheta})$$

- Treat $\boldsymbol{\eta}$ as unknowns to be sampled; likelihood simplifies.
- But problem scales with n (and m).

Marginal likelihood

$$\pi(\mathcal{Y} | \boldsymbol{\vartheta}) = \prod_{s=1}^n \mathcal{N}_p(\mathbf{y}_s; \boldsymbol{\mu}(\boldsymbol{\vartheta}), \boldsymbol{\Sigma}(\boldsymbol{\vartheta}))$$

where

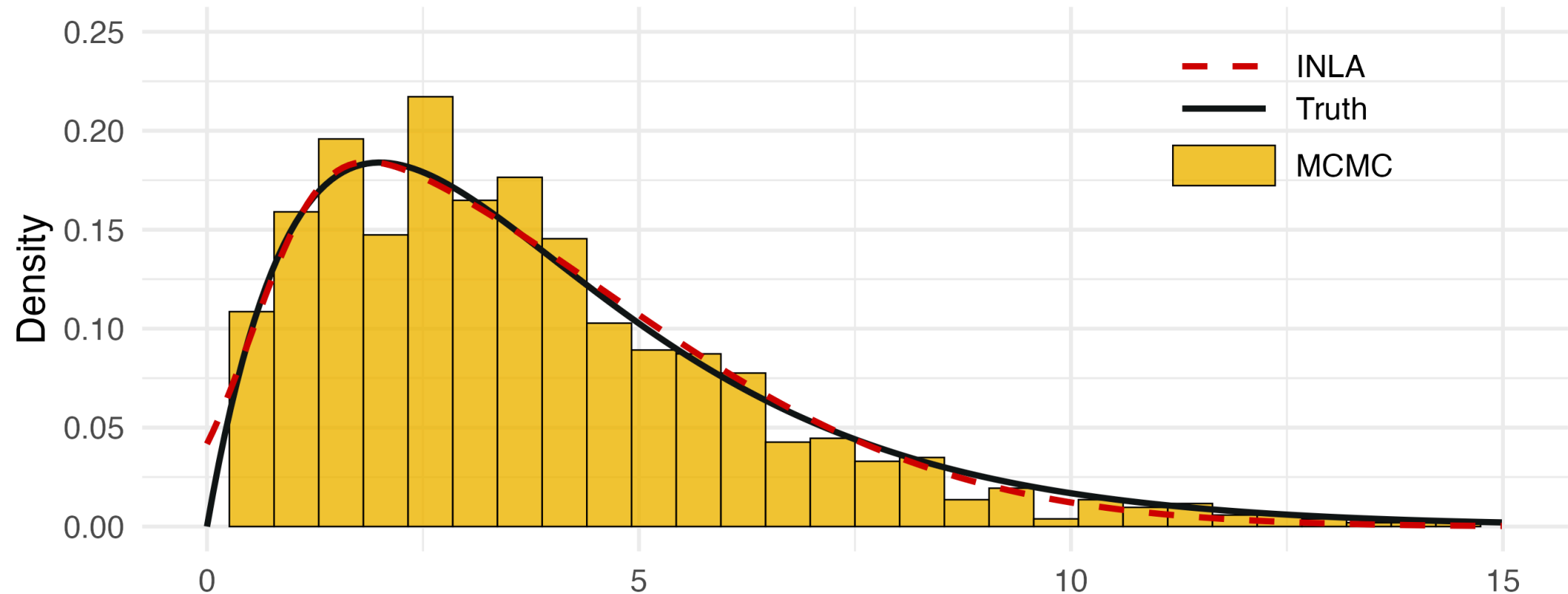
$$\begin{aligned} \boldsymbol{\mu}(\boldsymbol{\vartheta}) &= \boldsymbol{\nu} + \boldsymbol{\Lambda}(\mathbf{I} - \mathbf{B})^{-1}\boldsymbol{\alpha} \\ \boldsymbol{\Sigma}(\boldsymbol{\vartheta}) &= \boldsymbol{\Lambda}\boldsymbol{\Phi}\boldsymbol{\Lambda}^\top + \boldsymbol{\Theta} \\ \boldsymbol{\Phi} &:= (\mathbf{I} - \mathbf{B})^{-1}\boldsymbol{\Psi}(\mathbf{I} - \mathbf{B})^{-\top} \end{aligned}$$



MCMC-based estimation (*cont.*)

MCMC Sampling vs Approximate Inference

MCMC samples: 1000

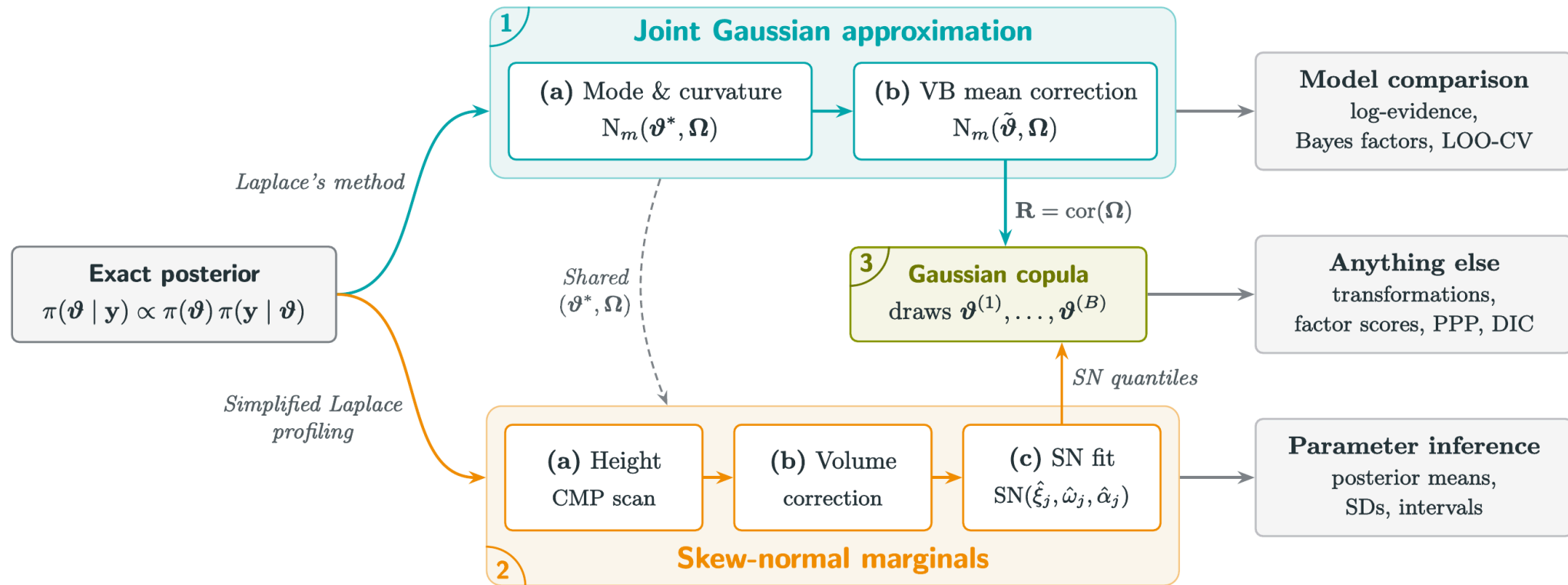


Many software choices to perform MCMC-based inference, e.g. **blavaan** (Merkle et al., 2021), **Stan** (Carpenter et al., 2017), **Mplus** (Muthén & Muthén, 2017).



INLA-based posterior approximation

Overview



- Not quite R-INLA: This is a ✨*bespoke*✨ INLA for SEM, making use of relevant key ideas (Laplace, skew-normal marginals, VB correction).
- Works because the exact posterior is cheap. For normal SEM, $\pi(\mathbf{y} | \boldsymbol{\vartheta})$ is closed form, so $\pi(\boldsymbol{\vartheta} | \mathbf{y})$ and its gradients are cheap to evaluate.



[Step 1a] Joint Gaussian approximation

💡 Laplace approximation (Kass & Raftery, 1995; Tierney et al., 1989)

Assume that $f(\boldsymbol{\vartheta}) := \log \{\pi(\mathbf{y} \mid \boldsymbol{\vartheta}) \pi(\boldsymbol{\vartheta})\}$ is twice continuously differentiable with a unique interior maximiser $\boldsymbol{\vartheta}^*$, at which $\mathbf{H} = -\nabla^2 f(\boldsymbol{\vartheta}^*)$ is positive definite. Consider the Taylor expansion of f about $\boldsymbol{\vartheta}^*$,

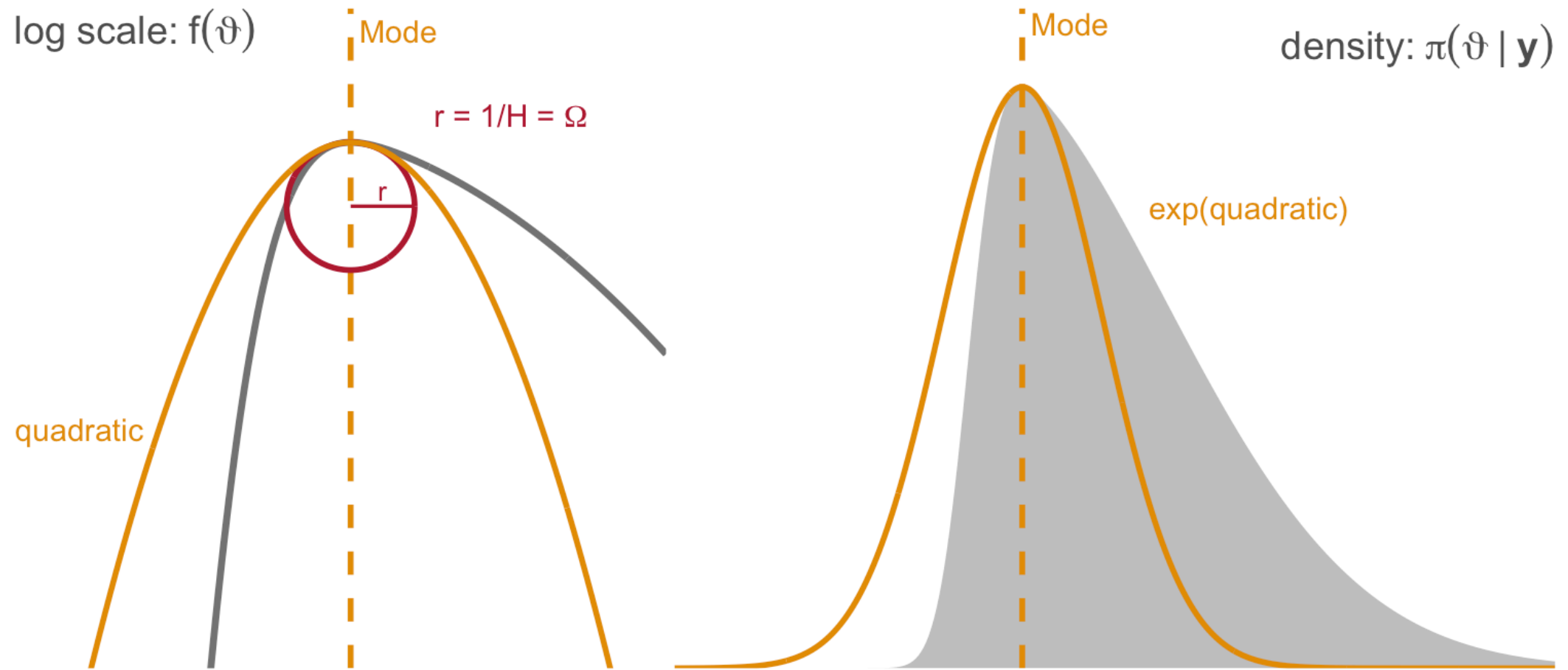
$$f(\boldsymbol{\vartheta}) \approx f(\boldsymbol{\vartheta}^*) - \frac{1}{2}(\boldsymbol{\vartheta} - \boldsymbol{\vartheta}^*)^\top \mathbf{H}(\boldsymbol{\vartheta} - \boldsymbol{\vartheta}^*). \quad (1)$$

Therefore, $\pi(\boldsymbol{\vartheta} \mid \mathbf{y}) \approx \mathcal{N}_m(\boldsymbol{\vartheta}; \boldsymbol{\vartheta}^*, \boldsymbol{\Omega})$, where $\boldsymbol{\Omega} = \mathbf{H}^{-1}$

- Work on an unconstrained scale (log variances, Fisher- z correlations). Make the log-posterior closer to quadratic, so Laplace is more accurate.
- Under standard regularity conditions, the approximation achieves a *relative* error of $O(n^{-1})$ for the posterior density and marginal likelihood (Rue et al., 2017) —c.f. additive $O(n^{-1/2})$ error typical of simulation-based methods.
- By the Bernstein-von Mises theorem, the posterior concentrates at a rate of $n^{-1/2}$ and converges to a Gaussian in total variation (van der Vaart, 1998).



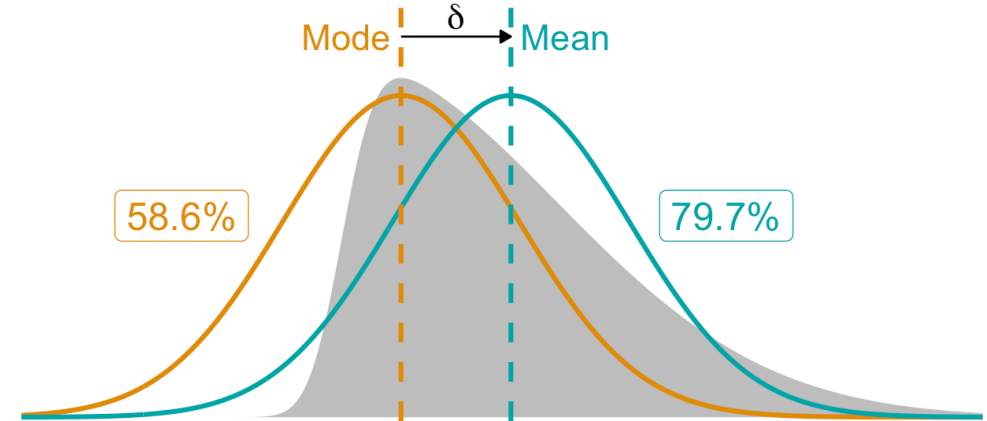
[Step 1a] Joint Gaussian approximation (*cont.*)





[Step 1b] Variational mean correction

- Posterior mode is often far away from the “typical set”, especially for skewed distributions.
- Laplace may misrepresent the bulk of the distribution—correction needed!



- van Niekerk & Rue (2024) propose a Variational Bayes (VB) correction to the mean of global Laplace:

$$\hat{\boldsymbol{\delta}} = \operatorname{argmin}_{\boldsymbol{\delta}} \mathbf{D}_{\text{KL}} \left(\mathcal{N}_m(\cdot; \boldsymbol{\vartheta}^* + \boldsymbol{\delta}, \mathbf{H}^{-1}) \parallel \pi(\cdot \mid \mathbf{y}) \right).$$

- Distribution that minimises *information loss* (Zellner, 1988).
- Reuse the Laplace curvature (good locally) and correct only the centre: an m -dimensional optimisation instead of re-estimating an $m \times m$ covariance.
- Shift the whole Gaussian towards the posterior mean.

Overlap (area shared by the two densities): $\int \min\{p(\boldsymbol{\vartheta}), q(\boldsymbol{\vartheta})\} d\boldsymbol{\vartheta} = 1 - \frac{1}{2} \int |p(\boldsymbol{\vartheta}) - q(\boldsymbol{\vartheta})| d\boldsymbol{\vartheta}$.

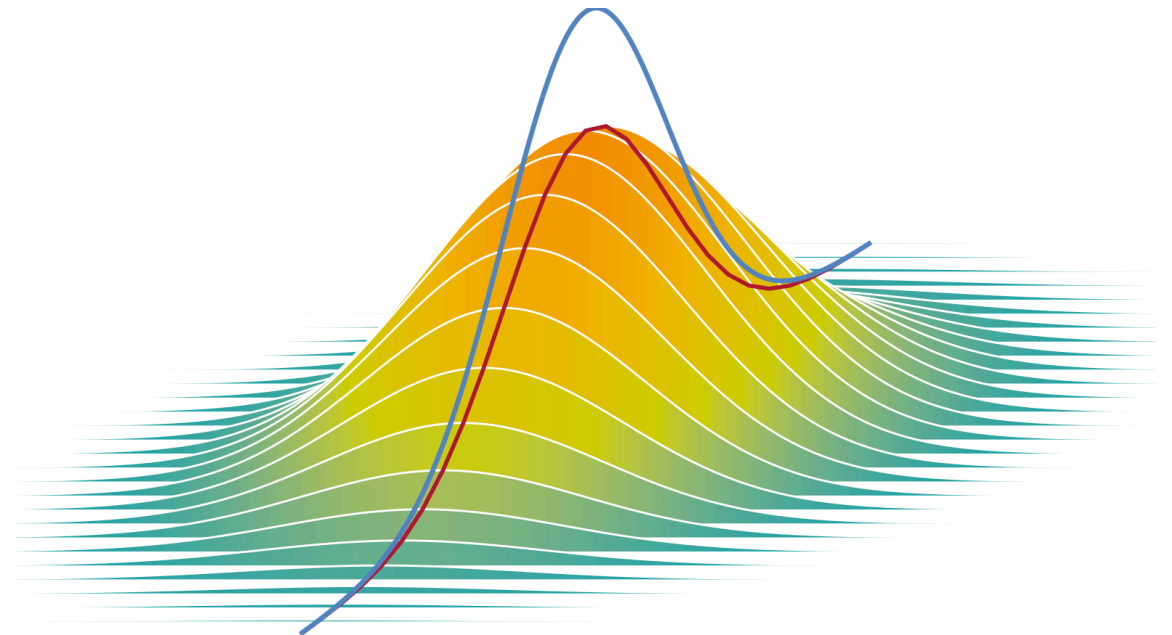


[Step 2] Posterior marginals

- Use **Laplace's method** (1), this time to approximate an integral:

$$\pi(\boldsymbol{\vartheta}_j \mid \mathbf{y}) = \int_{\mathbb{R}^{m-1}} e^{\log \pi(\boldsymbol{\vartheta}_j, \boldsymbol{\vartheta}_{-j} \mid \mathbf{y})} d\boldsymbol{\vartheta}_{-j} \approx \overbrace{\pi(\boldsymbol{\vartheta}_j, \boldsymbol{\vartheta}_{-j}^*(\boldsymbol{\vartheta}_j) \mid \mathbf{y})}^{\text{height}} \times \overbrace{(2\pi)^{\frac{m-1}{2}} |\mathbf{H}_{-j}(\boldsymbol{\vartheta}_j)|^{-1/2}}^{(\text{cond.}) \text{ width}}$$

- Geometric intuition: the marginal at each $\boldsymbol{\vartheta}_j$ is the volume of the posterior's slice there, roughly the slice's *peak height* times its *width*.
- This is **expensive** to compute, as for every $\boldsymbol{\vartheta}_j$ evaluation, need
 1. An optimisation of $\boldsymbol{\vartheta}_{-j} \in \mathbb{R}^{m-1}$ at this location; and
 2. The (log) determinant of the Hessian there.
- Next slides: two important results.





[Step 2a] Posterior marginals—the “height”

Lemma 1 (Conditional Mean Path) = No need for reoptimisation

Let $\boldsymbol{\vartheta} \sim \mathcal{N}_m(\boldsymbol{\vartheta}^*, \boldsymbol{\Omega})$ with $\boldsymbol{\Omega} \succ \mathbf{0}$. Then the marginal density of the j th component, $j = 1, \dots, m$, is proportional to the joint density evaluated along its CMP:

$$\pi(\vartheta_j = x) \propto \pi(\vartheta_j = x, \boldsymbol{\vartheta}_{-j} = \mathbf{E}[\boldsymbol{\vartheta}_{-j} \mid \vartheta_j = x]).$$

- Operationally,
 - i. Lay out a grid $\mathcal{G}_j = \{\boldsymbol{\vartheta}^* + t \mathbf{v}_j \mid t \in [-4, 4]\}$, $\mathbf{v}_j = \boldsymbol{\Omega}_{jj}^{1/2} \boldsymbol{\Omega}_{\cdot j}$.
 - ii. Evaluate joint posterior $\pi(\boldsymbol{\vartheta} \mid \mathbf{y})$ at each grid point.
- Scanning along the CMP is cheap; no re-optimisation needed.
- For Gaussian posteriors, this is **exact**; otherwise it provides a first-order approximation to a potentially curved ridge.



[Step 2b] Posterior marginals—the “width”

Lemma 2 (Efficient volume correction) = No need to factor dense Hessian

For the j th component along the CMP, $-\frac{1}{2}\log |\mathbf{H}_{-j}(t)| \approx \text{const.} + t\gamma'_j$. Then

$$\gamma'_j = -\frac{1}{2} \sum_{k=1}^m \frac{d}{dt} \left[\mathbf{L}_{\cdot k}^\top \frac{d}{d\mathbf{L}_{\cdot k}} \mathbf{g} \right]_{t=0} + \frac{1}{2} \frac{d}{dt} \left[\mathbf{v}_j^\top \frac{d}{d\mathbf{v}_j} \mathbf{g} \right]_{t=0},$$

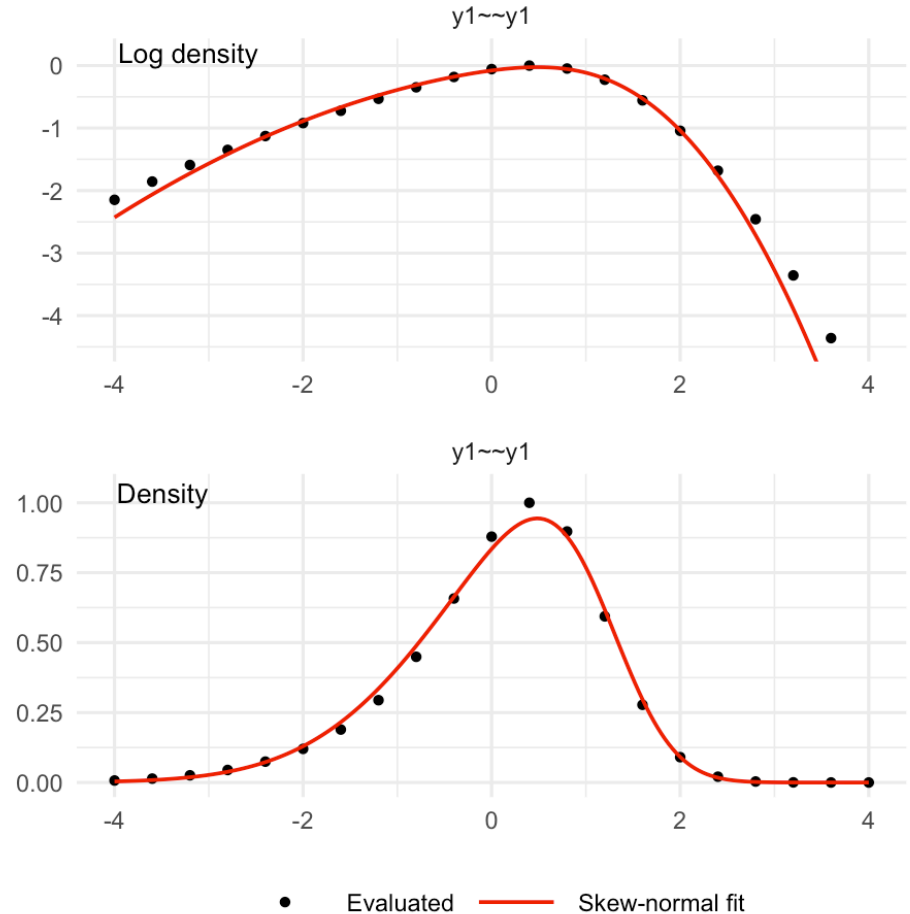
where $\mathbf{g}(\boldsymbol{\vartheta}) = -\nabla \log \pi(\boldsymbol{\vartheta} \mid \mathbf{y})$, and $\mathbf{H}(\boldsymbol{\vartheta}^*)^{-1} = \mathbf{L}\mathbf{L}^\top$.

- For non-symmetric densities, the conditional spread (“width”) changes along the scan line. This change can be computed entirely via directional derivatives.
- Forward difference cost is only $m + 2$ gradient evaluations for each j . Note: Baseline is unity in whitened coordinates.
 - i. [outer ∇^3] Shift to $\boldsymbol{\vartheta}^+$ along CMP, evaluate gradient.
 - ii. [inner ∇^2] Perturb along $\mathbf{L}_{\cdot k}$ and the scan direction $\mathbf{v}_j$, evaluating gradients.



[Step 2c] Posterior marginals—SN curve fit

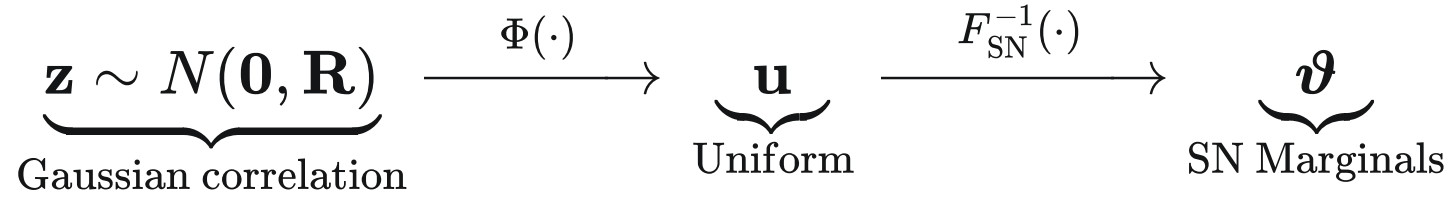
- Consider the paired points $\{\vartheta_j(t_k), h_{kj}\}_{k=1}^K$, where
 - $t_k \in [-4, 4]$ are standardised grid points,
 - $\vartheta(t_k) = \vartheta^* + t_k \mathbf{v}_j$ from the CMP, and
 - $h_{kj} := h_j(t_k) = \log \pi(\vartheta(t_k) \mid y) + t_k \gamma'_j$ are unnormalised ordinates.
- Fit a skew normal pdf to these points via weighted least squares. A convenient default is $w_{kj} \propto \exp(h_{kj})$, for $j = 1, \dots, m$.
- Map back from t to ϑ_j scale via the affine $\vartheta_j = \vartheta_j^* + t \sqrt{\Omega_{jj}}$.



$$(\hat{\xi}_j, \hat{\omega}_j, \hat{\alpha}_j, \hat{c}_j) = \operatorname{argmin}_{\xi, \omega, \alpha, c} \sum_{k=1}^K w_{kj} \left[h_{kj} - \{ \log f_{\text{SN}}(\vartheta_j(t_k); \xi, \omega, \alpha) + c \} \right]^2$$



[Step 3] Gaussian copula sampling



- Reconstruct joint posterior samples $\{\boldsymbol{\vartheta}^{(1)}, \dots, \boldsymbol{\vartheta}^{(B)}\}$ using a Copula (Nelsen, 2006), preserving Gaussian correlations:

$$\mathbf{R} = \mathbf{D}^{-1/2} \boldsymbol{\Omega} \mathbf{D}^{-1/2}, \quad \mathbf{D} = \text{diag}(\boldsymbol{\Omega}).$$

- This also allows us to derive approximate posteriors for
 - i. Parameter transformations, e.g. standardised solutions, non-linear transforms, etc.
 - ii. Posterior Predictive P-value (PPP), e.g. $\#\{T(\mathbf{S}_{\text{obs}}, \boldsymbol{\Sigma}(\boldsymbol{\vartheta})) > T(\mathbf{S}_{\text{rep}}, \boldsymbol{\Sigma}(\boldsymbol{\vartheta}))\}$.
 - iii. Factor scores, $\boldsymbol{\eta} \mid \{\mathcal{Y}, \boldsymbol{\vartheta}\} \sim N_q(\mathbf{m}(\boldsymbol{\vartheta}), \mathbf{V}(\boldsymbol{\vartheta}))$.
 - iv. Prediction and imputation.



R/INLAvaan

The R/INLAvaan package

lifecycle experimental

CRAN 0.3.2



```
1 # Check data set
2 glimpse(Kidney)
```

Rows: 250

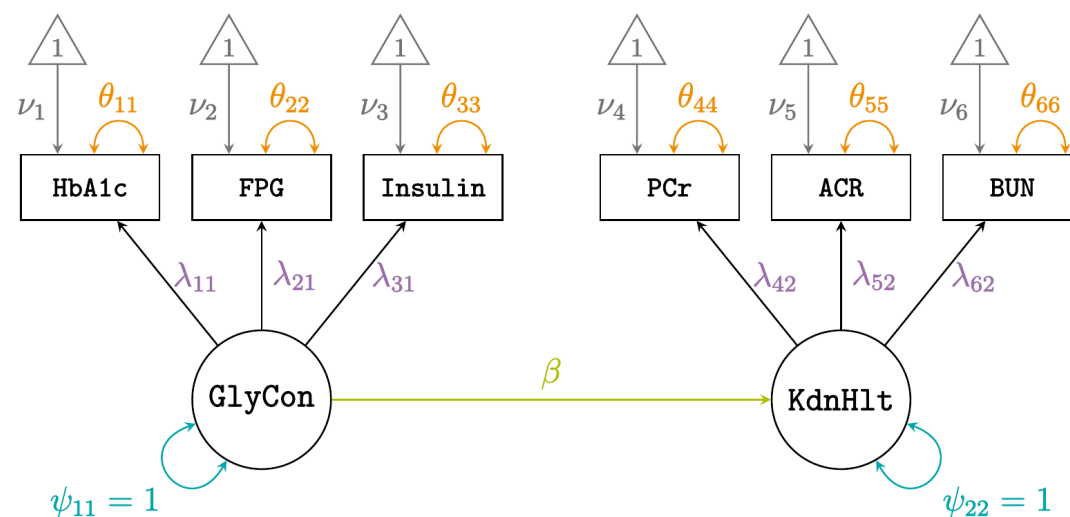
Columns: 6

```
$ y1 <dbl> 5.960822, 6.129040, 5.97609...
$ y2 <dbl> 4.767713, 6.616294, 5.64002...
$ y3 <dbl> 14.469238, 10.517574, 8.725...
$ y4 <dbl> 84.18186, 59.59659, 126.733...
$ y5 <dbl> 64.539671, 49.028101, 90.72...
$ y6 <dbl> 7.758444, 5.232205, 11.5583...
```

```
1 library(INLAvaan)
2 mod <- "
3   # Measurement model
4   GlyCon =~ y1 + y2 + y3
5   KdnHlt =~ y4 + y5 + y6
6
7   # Structural model
8   KdnHlt ~ GlyCon
9   "
```

```
i Mode finding and Hessian computation.
i Computing the Hessian.
✓ Posterior mode and Hessian. [44ms]
i Performing VB correction.
✓ VB correction; mean  $|\delta| = 0.085\sigma$ . [44ms]
∴ Fitting 0/19 skew-normal marginals.
✓ Fit 19/19 skew-normal marginals. [134ms]
∴ Posterior sampling and summarising.
✓ Summarise 1000 posterior draws. [224ms]
i Fit measures: PPP, DIC.
```

total
0.47 s





(b)lavaan S4 methods (Rosseel, 2012)

```
1 print(fit)
```

INLAvaan 0.3.2.9003 ended normally after 71 iterations

Estimator	BAYES
Optimization method	NLMINB
Number of model parameters	19
Number of observations	250

Model Test (User Model):

Marginal log-likelihood	-4029.054
PPP (Chi-square)	0.569

Also available:

- `fitmeasures()`
- `fitted()`
- `predict()`
- `standardisedsolution()`
- `vcov()`
- `plot()`

```
1 coef(fit)
```

GlyCon=~y1	GlyCon=~y2	GlyCon=~y3	KdnHlt=~y4	KdnHlt=~y5
1.090	1.161	5.027	21.610	19.651
KdnHlt=~y6	KdnHlt~GlyCon	y1~~y1	y2~~y2	y3~~y3
1.773	0.656	0.322	0.254	7.435
y4~~y4	y5~~y5	y6~~y6	y1~1	...
206.240	91.076	1.285	6.542	

Summary



```
1 summary(fit)
```

<output truncated>

Information Criteria:

Deviance (DIC)	7910.072
Effective parameters (pD)	19.026

Parameter Estimates:

Marginalisation method	SKEWNORM
VB correction	TRUE

Latent Variables:

	Estimate	SD	2.5%	97.5%	NMAD	Prior
GlyCon =~						
y1	1.090	0.063	0.972	1.218	0.005	normal(0,10)
y2	1.161	0.063	1.042	1.290	0.005	normal(0,10)
y3	5.027	0.292	4.476	5.625	0.005	normal(0,10)
KdnHlt =~						
v4	21.610	1.281	19.188	24.221	0.005	normal(0,10)





Comparison to MCMC

- Interchangeable commands (c.f. `bsem()`) with `blavaan` (Merkle et al., 2021) which interfaces with `Stan` (Carpenter et al., 2017).

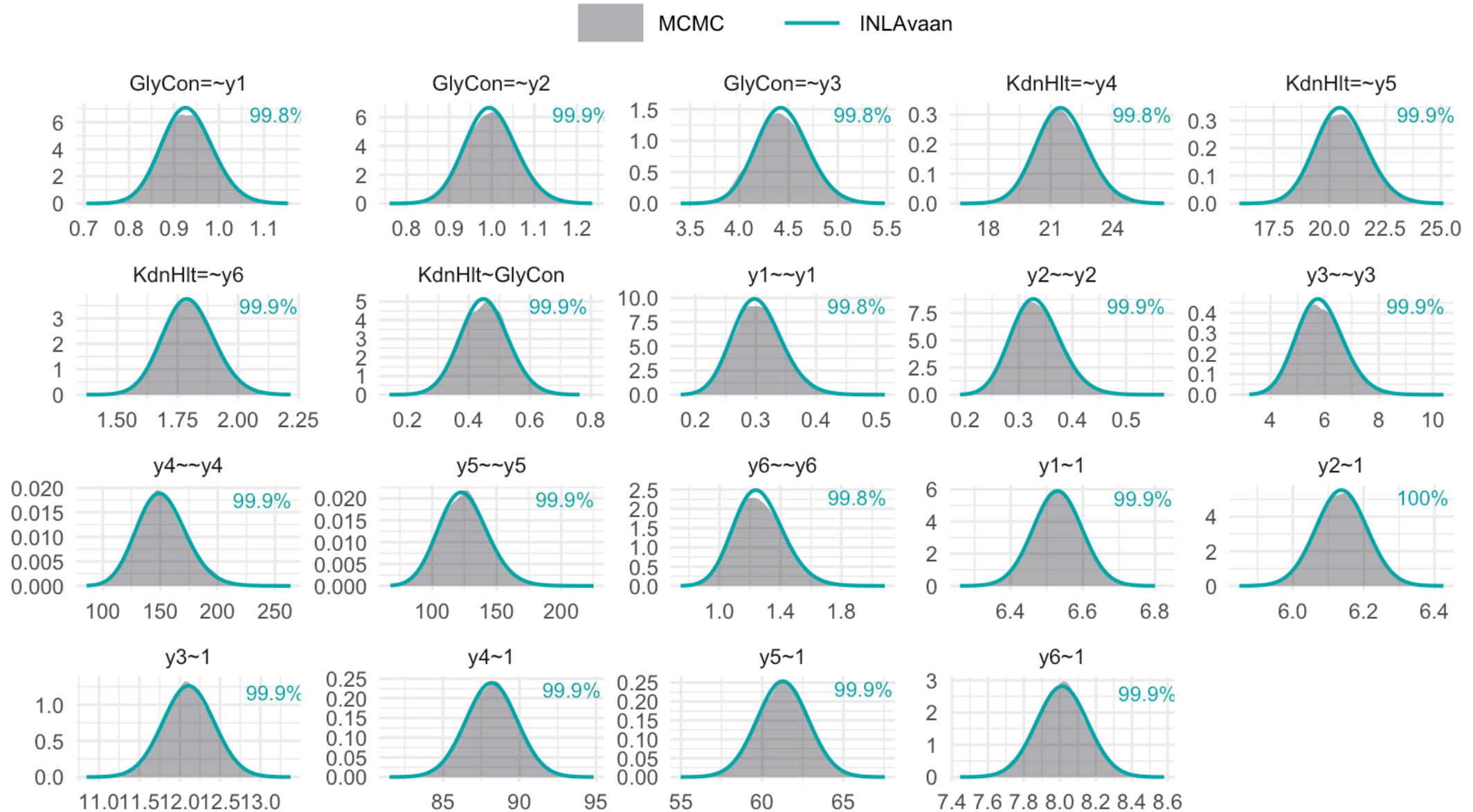
```
> library(blavaan)
Loading required package: Rcpp
This is blavaan 0.6-1
On multicore systems, we suggest use of future::plan("multicore") or
  future::plan("multisession") for faster post-MCMC computations.
>
> fit_blav <- bsem(mod, Kidney, meanstructure = TRUE, std.lv = TRUE)

SAMPLING FOR MODEL 'stanmarg' NOW (CHAIN 1).
Chain 1:
Chain 1: Gradient evaluation took 0.000331 seconds
Chain 1: 1000 transitions using 10 leapfrog steps per transition would take 3.31 seconds.
Chain 1: Adjust your expectations accordingly!
Chain 1:
Chain 1:
Chain 1: Iteration:    1 / 1500 [  0%] (Warmup)
Chain 1: Iteration:  150 / 1500 [ 10%] (Warmup)
```





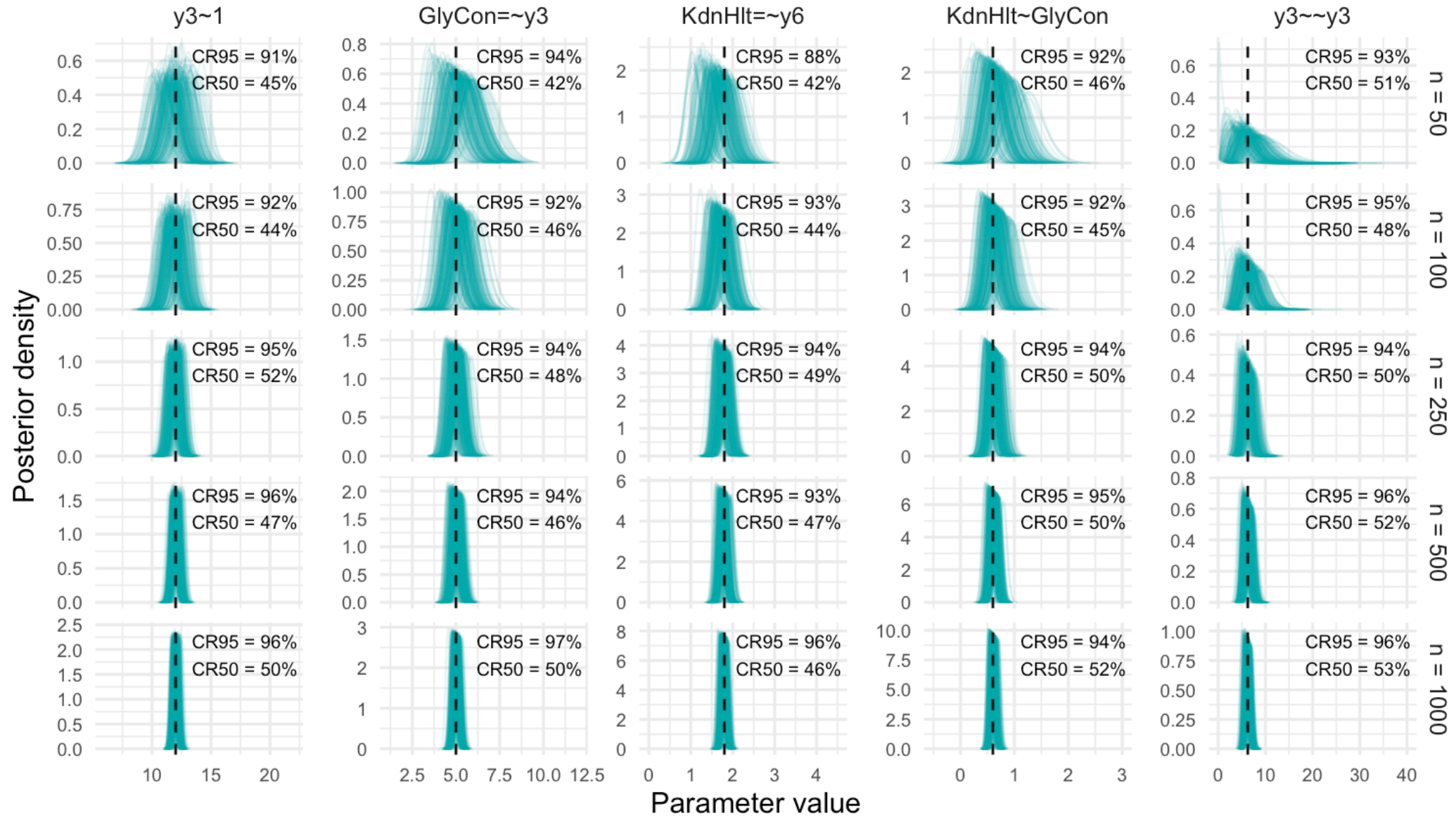
Comparison to MCMC (cont.)



MCMC: 3 chains of 1,000 draws each, after 500 burn-in. % are 1 minus Jensen-Shannon divergences.



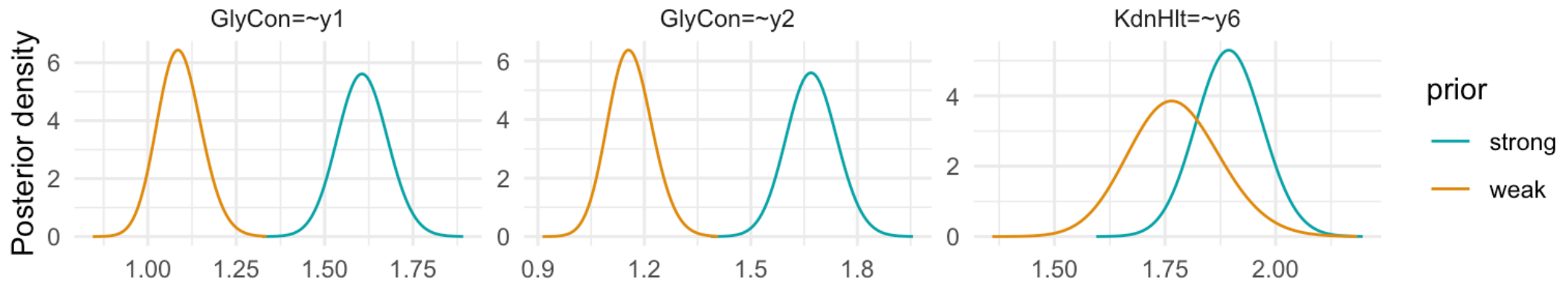
Parameter recovery from simulated data



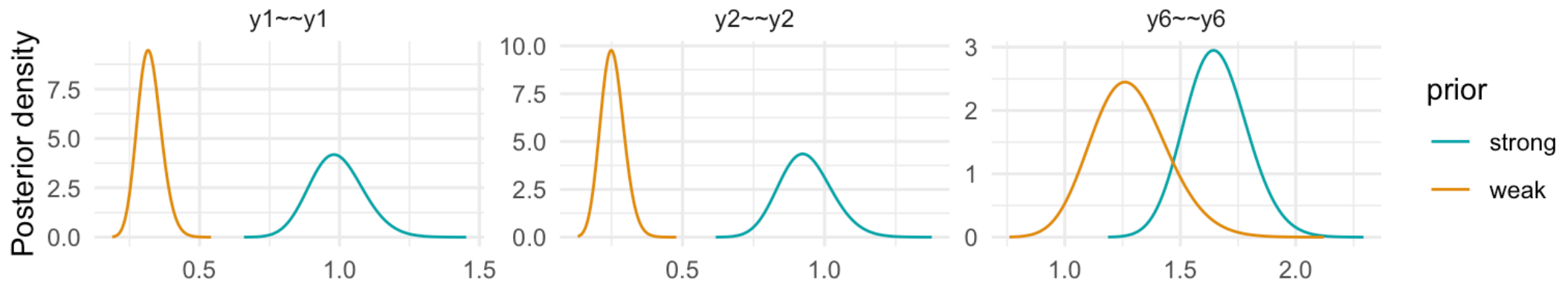


Effects of prior distributions

- Loadings: $\lambda \sim N(0, 10^2) \mapsto \lambda \sim N(2, 0.1^2)$



- Residual variances: $\theta^{1/2} \sim \Gamma(1, 0.5) \mapsto \theta \sim \Gamma(100, 50)$





Factor scores and latent regression

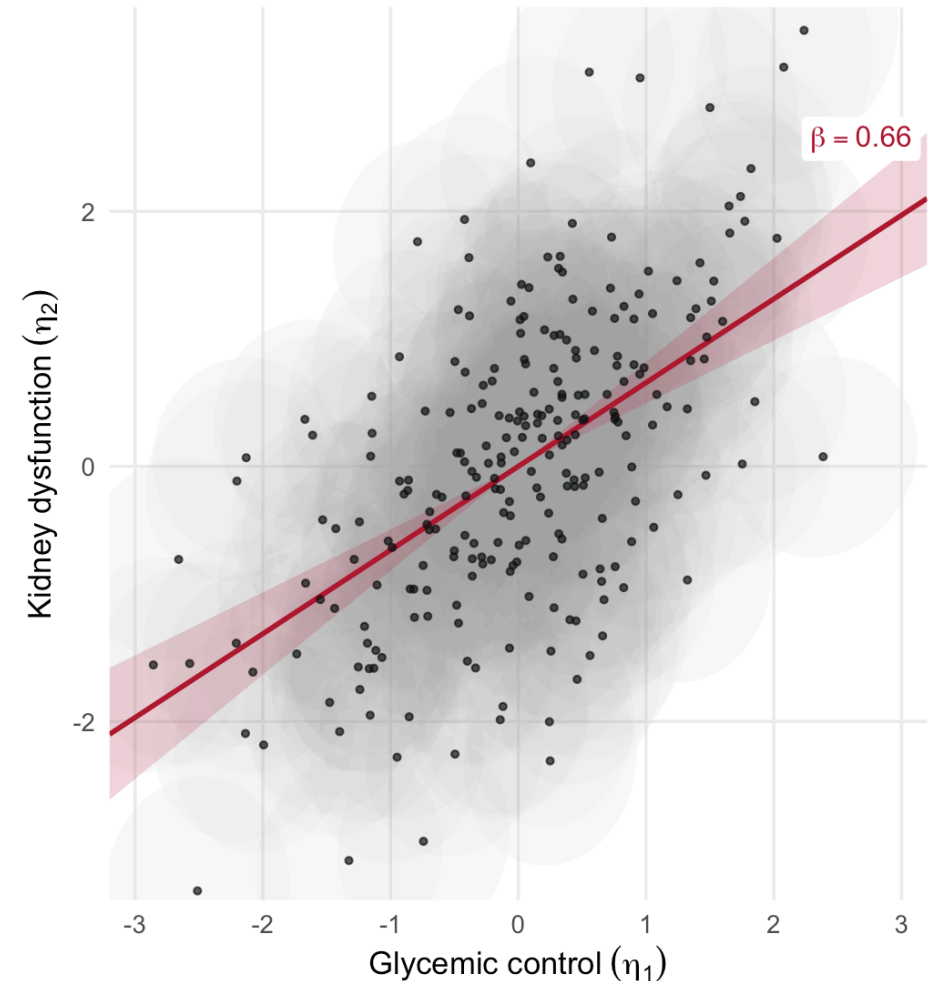
```
1 eta <- predict(fit, type = "lv")
2 print(eta)
```

First sample:

	GlyCon	KdnHlt
1	-0.370	-0.0255
2	-0.274	-1.1595
3	0.253	1.6439
4	1.311	-1.8157
5	0.431	1.0514

i 245 more rows
i Use `summary()` to see summary statistics

- Posterior of η_i for all $n = 250$ patients.
- **Halo**: 95% posterior region (SD ≈ 0.28 for η_1 , 0.33 for η_2).
- **Line**: SEM's $\beta = 0.66$ [0.49, 0.82], estimated *jointly*.





Refit-free Leave-One-Out Cross-Validation

```
1 tictoc::tic()
2 loo(fit)
```

— Leave-one-subject-out ————— 250 subjects, second-order —

	Estimate	SE
elpd_loo	-3955.0	28.7
p_loo	18.8	1.3
looic	7909.9	57.3

— Curvature check —————

first-to-second-order gap	9.7
pD/2 (trace)	9.3
excess over pD/2 (trace)	+4.7%

- i The gap approaches pD/2 (trace) from above. A large excess says the second-order expansion has not settled over the sample.

```
1 tictoc::toc()
```

0.037 sec elapsed

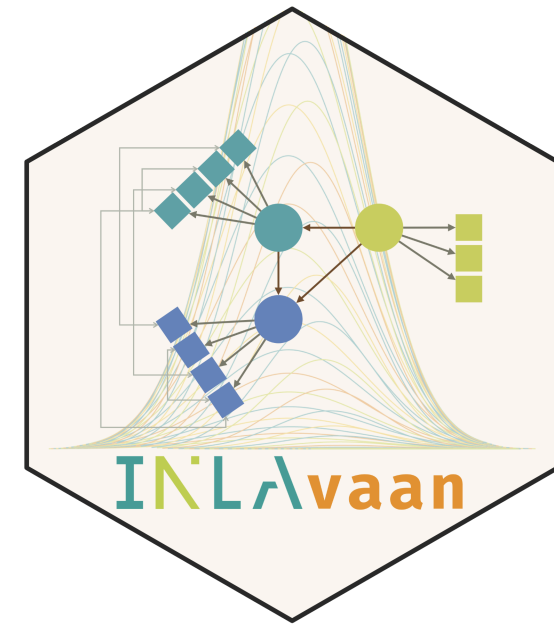


Conclusion



Summary

- 🚀 INLA methodology offers a rapid alternative for SEM, delivering Bayesian results at near the speed of frequentist estimators.
 - Not aiming to replace MCMC; use case is model development, testing, and exploratory analysis.
- Forward looking
 - Non-Gaussian outcomes (e.g. categorical, count)
 - Better prior support (priors for correlation matrices [Freni-Sterrantino et al., 2026](#); penalised complexity priors [Simpson et al., 2017](#))
 - Dynamic SEM, including continuous-time SEM ([Driver et al., 2017](#))
 - Spatiotemporal extensions ([Krainski et al., 2018](#))
- Vignettes and tutorials available on <https://inlavan.haziqj.ml/>: Equality constraints, defined parameters, multigroup, multilevel, and missing data.





شكراً جزيلًا

<https://haziqj.ml/kaust-gradsem>



References

- Burnell, R., Hao, H., Conway, A. R. A., & Orallo, J. H. (2023). *Revealing the structure of language model capabilities*. <https://doi.org/10.48550/arXiv.2306.10062>
- Cafiero, C., Viviani, S., & Nord, M. (2018). Food security measurement in a global context: The food insecurity experience scale. *Measurement*, 116, 146–152. <https://doi.org/10.1016/j.measurement.2017.10.065>
- Carpenter, B., Gelman, A., Hoffman, M. D., Lee, D., Goodrich, B., Betancourt, M., Brubaker, M., Guo, J., Li, P., & Riddell, A. (2017). *Stan*: A probabilistic programming language. *Journal of Statistical Software*, 76(1), 1–32. <https://doi.org/10.18637/jss.v076.i01>
- Driver, C. C., Oud, J. H. L., & Voelkle, M. C. (2017). Continuous Time Structural Equation Modeling with R Package ctsem. *Journal of Statistical Software*, 77(5). <https://doi.org/10.18637/jss.v077.i05>
- European Social Survey European Research Infrastructure (ESS ERIC). (2023). *ESS5 - integrated file, edition 3.5 (Austria not included)* (Version 1) [Data set]. Sikt - Norwegian Agency for Shared Services in Education and Research. https://doi.org/10.21338/ESS5E03_5
- Freni-Sterrantino, A., van Niekerk, J., Krainski, E., Rustand, D., Jamil, H., & Rue, H. (2026). *Informative distance-based priors for correlation matrices centred on a target reference*. <https://doi.org/10.48550/arXiv.2607.18926>
- Grace, J. B., Anderson, T. M., Seabloom, E. W., Borer, E. T., Adler, P. B., Harpole, W. S., Hautier, Y., Hillebrand, H., Lind, E. M., Pärtel, M., Bakker, J. D., Buckley, Y. M., Crawley, M. J., Damschen, E. I., Davies, K. F., Fay, P. A., Firn, J., Gruner, D. S., Hector, A., ... Smith, M. D. (2016). Integrative modelling reveals mechanisms linking productivity and plant species richness. *Nature*, 529(7586), 390–393. <https://doi.org/10.1038/nature16524>
- Jackson, J., Bradford, B., Hough, M., Kuha, J., Stares, S., Widdop, S., Fitzgerald, R., Yordanova, M., & Galev, T. (2011). Developing European indicators of trust in justice. *European Journal of Criminology*, 8(4), 267–285. <https://doi.org/10.1177/1477370811411458>
- Jackson, J., Hough, M., Bradford, B., Pooler, T., Hohl, K., & Kuha, J. (2011, November 1). *Trust in justice: Topline results from the round 5 of the european social survey* (SSRN Scholarly Paper No. 2231094). <https://papers.ssrn.com/abstract=2231094> (Pre-published)
- Kass, R. E., & Raftery, A. E. (1995). Bayes factors. *Journal of the American Statistical Association*, 90(430), 773–795. <https://doi.org/10.1080/01621459.1995.10476572>
- Krainski, E., Gómez-Rubio, V., Bakka, H., Lenzi, A., Castro-Camilo, D., Simpson, D. P., Lindgren, F., & Rue, H. (2018). *Advanced spatial modeling with stochastic partial differential equations using R and INLA*. Chapman and Hall/CRC. <https://doi.org/10.1201/9780429031892>
- Merkle, E. C., Fitzsimmons, E., Uanhoro, J., & Goodrich, B. (2021). Efficient Bayesian structural equation modeling in Stan. *Journal of Statistical Software*, 100(6), 1–22. <https://doi.org/10.18637/jss.v100.i06>
- Muthén, B., & Muthén, L. (2017). *Mplus user's guide version 8.1* (Version 8) [Computer software]. <https://www.statmodel.com/>
- Nelsen, R. B. (2006). *An introduction to copulas* (2nd ed.). Springer.
- Rosseel, Y. (2012). lavaan: An R package for structural equation modeling. *Journal of Statistical Software*, 48(2), 1–36. <https://doi.org/10.18637/jss.v048.i02>
- Rue, H., Martino, S., & Chopin, N. (2009). Approximate Bayesian inference for latent Gaussian models by using integrated nested Laplace approximations. *Journal of the Royal Statistical Society: Series B (Statistical Methodology)*, 71(2), 319–392. <https://doi.org/10.1111/j.1467-9868.2008.00700.x>
- Rue, H., Riebler, A., Sørbye, S. H., Illian, J. B., Simpson, D. P., & Lindgren, F. (2017). Bayesian computing with INLA: A review. *Annual Review of Statistics and Its Application*, 4(1), 395–421. <https://doi.org/10.1146/annurev-statistics-060116-054045>
- Simpson, D. P., Rue, H., Riebler, A., Martins, T. G., & Sørbye, S. H. (2017). Penalising model component complexity: A principled, practical approach to constructing priors. *Statistical Science*, 32(1), 1–28. <https://doi.org/10.1214/16-STS576>
- Song, X.-Y., & Lee, S.-Y. (2012). *Basic and advanced Bayesian structural equation modeling: With applications in the medical and behavioral sciences*. Wiley. <https://doi.org/10.1002/9781118358887>
- Tierney, L., Kass, R. E., & Kadane, J. B. (1989). Fully exponential Laplace approximations to expectations and variances of nonpositive functions. *Journal of the American Statistical Association*, 84(407), 710–716. <https://doi.org/10.1080/01621459.1989.10478824>
- van der Vaart, A. W. (1998). *Asymptotic statistics*. Cambridge University Press. <https://doi.org/10.1017/CBO9780511802256>
- van Niekerk, J., & Rue, H. (2024). Low-rank variational Bayes correction to the Laplace method. *Journal of Machine Learning Research*, 25(62), 1–25. <https://jmlr.org/papers/v25/21-1405.html>
- Young, S. L., Boateng, G. O., Jamaluddine, Z., Miller, J. D., Frongillo, E. A., Neilands, T. B., Collins, S. M., Wutich, A., Jepson, W. E., & Stoler, J. (2019). The Household Water InSecurity Experiences (HWISE) Scale: Development and validation of a household water insecurity measure for low-income and middle-income countries. *BMJ Global Health*, 4(5), e001750. <https://doi.org/10.1136/bmjgh-2019-001750>
- Zellner, A. (1988). Optimal information processing and Bayes's theorem. *The American Statistician*, 42(4), 278–280. <https://doi.org/10.2307/2685143>